

SHUNAN SHENG

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 Social Work Building, 1255 Amsterdam Avenue, New York, NY 10027

EDUCATION

Ph.D. in Statistics, Columbia University *Sept 2022 - Present*
Supported by Bloomberg Quantitative Finance Ph.D. Fellowship 2024 - present
Advisor: Prof. Marcel Nutz

B.S. in Mathematical Sciences, Nanyang Technological University, Singapore *Aug 2018 - May 2022*
Awarded the Senior Middle 2 Scholarship
Advisor: Prof. Ariel Neufeld

RESEARCH INTERESTS

- Optimal Transport, Robust Finance, Variational Inference, Sampling

PUBLICATIONS & PREPRINTS

González-Sanz, A., Mordant, G., & **Sheng, S.** (2026). Empirical Optimal Transport Potentials: Fast Rates and a Functional Central Limit Theorem. arXiv preprint arXiv:2608.00649.

González-Sanz, A., **Sheng, S.**, Wu, B., & Avella Medina, M. (2026). The Influence Function of Transport-based Quantiles. arXiv preprint arXiv:2607.19080.

Liang, B., Nutz, M., **Sheng, S.**, & Tissot-Daguette, V. (2026). Bid-Ask Martingale Optimal Transport. arXiv preprint arXiv:2603.24605. (*Major revision at Finance and Stochastics*)

Sheng, S., Wu, B., Zhu, B., Chewi, S., & Pooladian, A. A. (2025). Theory and Computation for Structured Variational Inference. arXiv preprint arXiv:2511.09897.

Sheng, S., Wu, B., & González-Sanz, A. (2025). Mode Collapse of Mean-Field Variational Inference. arXiv preprint arXiv:2510.17063.

Sheng, S., Wu, B., González-Sanz, A., & Nutz, M. (2025). Stability of Mean-Field Variational Inference. arXiv preprint arXiv:2506.07856. (*Major revision at Probability Theory and Related Fields*)

González-Sanz, A., & **Sheng, S.** (2024). Linearization of Monge-Ampère Equations and Statistical Applications. arXiv preprint arXiv:2408.06534.

Sheng, S., Xiang, Q., Nevat, I., & Neufeld, A. (2024). Binary spatial random field reconstruction from non-Gaussian inhomogeneous time-series observations. Journal of the Franklin Institute, 361(2), 612-636.

Ren, J., Yu, C., **Sheng, S.**, Ma, X., Zhao, H., Yi, S., & Li, H. (2020). Balanced meta-softmax for long-tailed visual recognition. Advances in Neural Information Processing Systems, 33, 4175-4186.

INDUSTRY RELATED EXPERIENCE

Bloomberg L.P. *Summers 2025, 2026*
Quantitative Researcher, Internship

National University of Singapore, Singapore *May 2021 - Aug 2021*
Undergraduate Researcher (Research in Industrial Projects for Students, RIPS 2021)

SenseTime Research, Singapore *Nov 2019 - Jun 2020*
Algorithm Researcher, Internship

AWARDS & HONORS

<i>Quantitative Finance Ph.D. Fellowship</i> , Bloomberg L.P., New York	<i>Sept 2024 - Present</i>
<i>Singapore National Academy of Science (SNAS) Award</i> , Singapore	<i>Jun 2022</i>
<i>Senior Middle 2 Scholarship</i> , NTU & Ministry of Education (MOE), Singapore	<i>Aug 2017 - Jun 2022</i>
<i>Stonehage Fleming/Marais Prize</i> , Simon Marais Mathematical Competition	<i>Dec 2019</i>

SERVICE

Reviewer: Journal of the American Statistical Association, SIAM Journal on Optimization, SIAM Journal on Mathematics of Data Science, ICLR 2026, Statistics and Probability Letters, Computational Statistics and Data Analysis

Organizer: Minghui Yu Memorial Conference 2025

TEACHING

Teaching Assistant at Columbia University

Time Series Modeling (Master/undergrad-level, GR5263/GU4263)	<i>Fall 2024</i>
Nonparametric Statistics (Master-level, GR5222)	<i>Spring 2024</i>
Introduction to Statistics with Calculus (Undergrad-level, UN1201)	<i>Fall 2023</i>
Introduction to Statistical Reasoning (Undergrad-level, UN1001)	<i>Summer 2023</i>
Statistical Machine Learning (Master-level, GR5241)	<i>Spring 2023</i>
Applied Linear Regression (Undergrad-level, UN2103)	<i>Fall 2022</i>

TALKS & POSTERS

ICORS 2026 International Conference on Robust Statistics (invited talk)	<i>July 2026</i>
CIB Data Science Seminar (virtual), Wells Fargo Corporate & Investment Banking	<i>May 2026</i>
Workshop on Robust Statistics (poster), Columbia University	<i>April 2026</i>
Minghui Yu Memorial Conference 2026 (contributed talk), Columbia University	<i>April 2026</i>
Bloomberg Quant (BBQ) Seminar (lightning talk), Bloomberg L.P.	<i>Mar 2026</i>
ICSIDS 2025 International Conference on Statistics and Data Science (contributed talk)	<i>Dec 2025</i>
Minghui Yu Memorial Conference 2025 (lightning talk), Columbia University	<i>April 2025</i>
Optimization and Statistical Learning Workshop Workshop (poster), Columbia University	<i>April 2025</i>
Statistics and Optimal Transport Workshop (poster), Columbia University	<i>Mar 2025</i>
ICSIDS 2024 International Conference on Statistics and Data Science (poster)	<i>Dec 2024</i>
FDS Conference: Recent Advances and Future Directions for Sampling (poster), Yale University	<i>Oct 2024</i>